CSE470: Software Engineering

Software Requirements Specification (SRS)

Project Title

Prepared By:

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Project Information:

Section	11
Section Faculty	Nazmus Sakib Touhid
GitHub Repository Link	https://github.com/tahmidislam2-star/Medisync
Probable Tech Stack	React, Fast API, MySQL

1. Introduction

1.1 Purpose

MediSync is a comprehensive healthcare platform designed for Bangladesh, enabling users to manage medical records, access test reports and prescriptions, find hospitals and doctors, schedule appointments, and receive online consultations. For doctors, MediSync simplifies appointment management, patient history review, messaging, and telehealth services. By connecting patients and healthcare providers in one platform, MediSync makes healthcare more accessible, convenient, and efficient.

1.2 Scope

MediSync minimizes the hassle patients have to go through to find hospital and doctors according to their needs. Meanwhile, Doctors can keep track of patient reports in one go. MediSync aims to centralize healthcare services in Bangladesh by connecting patients, doctors, hospitals, and diagnostic centers through a unified digital platform.

2. Overall Description

2.1 Product Perspective

While most healthcare apps in Bangladesh focus on individual hospitals or telemedicine, MediSync brings the entire healthcare ecosystem together, seamlessly connecting patients, doctors, and hospitals on one unified platform.

2.2 Product High-Level Summary

- *Secure Accounts, Roles, and Privacy*: Users register and log in with Google account, and role-based access control limits each role to what it is permitted to see.

- *Doctor and Hospital Discovery with Transparent Rankings:* Patients search and filter doctors and hospitals by specialty, location, fee, availability, rating, and distance, and view detailed profile pages.
- *Appointments, Payments, and Telehealth:* Patients book real-time slots, pay consultation fees online, and reschedule or cancel.
- *Lab Tests and Results Management*: Doctors order lab tests, and each test is tracked.

2.3 Users and Characteristics

- Patient: Search, book, view reports, message, review.
- Doctor: Manage schedule, view patient history, reply to messages, run telehealth sessions.
- Hospital Admin: Upload test results, manage the hospital profile, verify doctors and hospitals, departments, and lab turnaround data.
- System Admin: Moderate reviews, manage the platform.

2.4 Operating Environment

MediSync is a web app. It runs in any modern browser.

3. System Architecture

3.1 Technology Stack

Frontend: React.

Backend: Fast API.

4. Requirements and Features

Requirement ID	Requirement	Feature Description
1	User Registration	Patients can login/sign up with Google.
2	Role Based Access	Hospital admins register doctors to their hospitals.
3	Patient Profile and Medical History	Patients can view and change their personal information. They can also view their previous medical reports such as prescriptions and test results.
4	Doctor Search	Users can search doctor by specialty, location, fee, and availability.
5	Hospital Filtering	Users can filter hospitals by ranking, department, cost, and distance.
6	Doctor and Hospital Profile Page	Doctors and hospital can manage their profile by editing/ adding their information.
7	Appointment Booking with Real-time Slots	Patients can book their preferred slots for doctor appointment.
8	Reschedule and Cancel Appointments	Patients can reschedule or cancel their appointments if needed.

9	Real-time and Video Calling Between Doctors and Patients	Patients can consult with their preferred doctors online by text messages or video calls.
10	Online Payment for Telemedicine Service	Patients can pay for telemedicine services and book slots.
11	Lab Test Ordering and Tracking	Patients can track their lab tests (Ordered → Processing → Ready) and also book lab tests in their preferred hospital.
12	Secure Lab Result Downloading	Patients can download lab report as pdf file and save it in their local device.
13	E-Prescription generation	Doctors can generate prescriptions for individual patients.
14	Report Comparison Overtime	Patients can compare their current lab reports with their past reports.
15	User Removal and Workflow	Hospital admins can ban/ remove specific doctors or patients.
16	Hospital Service Information	Users can see what services a hospital provides.
17	Hospital and Doctor Rating	Verified users can rate hospitals and doctors.
18	Doctor Profile view	Doctors can view and track their patients.

19	Review Moderation	System Admin can manage reviews.
20	Hospital Access Control	System manager can register a hospital and add hospital admin.

5. Non-Functional Requirements (Optional)

Category	Requirement
Performance	User messages are delivered in real-time.
Security	Sensitive health data is encrypted.